Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable Iridium-Free Catalyst-Coated Membrane Electrode Assemblies for Cost-Efficient PEM Water Electrolysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (**x**), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Organic/Inorganic Material & Catalyst Manufacturing Technologies	Catalyst Application Technologies	New & Renewable Energy	Hydrogen / Fuel Cells
Concept of Proposed Technology	The proposed technology focuses on the development and scalable manufacturing of catalyst-coated membrane electrode assemblies (MEAs) for PEM water electrolysis, with a strong emphasis on eliminating iridium-based oxygen evolution catalysts and significantly reducing noble-metal content. **The core innovation lies in combining:** • **Iridium-free oxygen evolution reaction (OER) catalyst systems** based on earth-abundant transition metal oxides, • **Low-platinum hydrogen evolution reaction (HER) electrodes**, and • **Industrial-compatible MEA manufacturing processes**, suitable for continuous or roll-to-roll production. The technology targets a **substantial reduction of material and manufacturing costs** while maintaining high efficiency, durability, and scalability of PEM electrolyzers.			

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Rationale for Joint R&D Project	Current PEM electrolysis technology is strongly constrained by the high cost and limited availability of iridium, which represents a major barrier to large-scale deployment of green hydrogen systems. Replacing iridium without compromising performance requires advanced materials development, catalyst synthesis, and manufacturing know-how, as well as systematic electrochemical validation. **A joint R&D project enables:** • Integration of **material synthesis expertise**, **electrochemical characterization**, and **manufacturing technology**, • Development of **application-oriented solutions** aligned with industrial requirements, and • Acceleration of **technology readiness** toward pilot-scale and commercial relevance. → This collaboration is particularly attractive for **Korean SMEs** seeking next-generation hydrogen technologies with strong cost-reduction potential.
Key Words	1. PEM Electrolysis 2. Iridium-Free Catalysts 3. Catalyst-Coated Membranes 4. MEA Manufacturing 5. Green Hydrogen

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	TRL 3–4 (laboratory validation of materials and components)	End of the Project	TRL 6 (prototype MEAs validated at short-stack level)
R&D Objectives	1. Develop iridium-free OER catalyst systems with competitive activity and long-term stability. 2. Reduce noble-metal loading on the HER side while maintaining high efficiency. 3. Establish scalable manufacturing processes for catalyst-coated membranes and MEAs. 4. Validate electrochemical performance, durability, and scalability from single-cell to short-stack level.			
R&D Target Specifications	• ≥ 30–50 % reduction in catalyst material cost compared to state-of-the-art iridium-based MEAs • Stable operation at industrially relevant current densities • Demonstrated durability under dynamic operating conditions • Reproducible MEA fabrication on areas ≥ 100 cm²			
R&D Plan	**Year 1 – Materials and Catalyst Development** • Synthesis and optimization of iridium-free OER catalyst materials • Optimization of low-platinum HER catalyst layers • Electrochemical screening and benchmarking at laboratory scale **Year 2 – MEA Fabrication and Process Scale-Up** • Transfer of optimized catalysts into catalyst-coated membrane (CCM) concepts • Development of scalable deposition and lamination processes • Single-cell performance and durability testing **Year 3 – Prototype Validation and System Relevance** • Fabrication of large-area MEAs • Validation in short-stack configurations • Assessment of degradation behavior and manufacturing robustness			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Design and synthesis of catalyst materials • Development of MEA manufacturing processes • Electrochemical testing and performance evaluation • Data analysis and optimization of materials and processes
Expected R&R of Korean SME Partner	• Contribution to manufacturing process development and scale-up • Prototype fabrication and engineering support • Validation under application-relevant conditions • Preparation for commercialization and industrial integration

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Demonstration of iridium-free PEM electrolysis MEAs with industrial relevance • Significant reduction of material-related system costs • Improved scalability and supply-chain resilience for hydrogen technologies • Strong foundation for commercialization-oriented follow-up projects

6. References

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization